UNIVERSITY OF BUEA
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FACULTY OF SCIENCE
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DEPARTMENT OF COMPUTER SCIENCE

COURSE OUTLINE FOR IST608 (IAdvanced Topics in Database Management) – Second Semester 2023/2024
COURSE TITLE: Advanced Topics in Database Management
CREDIT VALUE: 6
NAMES OF COURSE INSTRUCTORS Dr Denis L. Nkweteyim
OFFICE HOURS:

WEEK	SLOT		TOPICS	ACTIVITY		
	Day	Time		L	T	P
1	1		Introduction to Transaction Processing Concepts and Theory	√		
	2		Transaction Processing I: Introduction to Transaction, Transaction and System Concepts, Desirable Properties of Transactions	√		
2	3		Transaction Processing II: Characterizing Schedules based on Recoverability, Characterizing Schedules based on Serializability, Transaction Support in SQL	√		
	4		Database Recovery Techniques: Purpose of Database Recovery, Types of Failure, The Log File, Data Cache and Buffers, Data Updating, Roll-back (Undo) and Roll-Forward (Redo), Checkpointing, Recovery Techniques, Recovery in Multidatabase Transactions	√		
3	5		Database Security I: Introduction to Database Security Issues, Discretionary Access Control Based on Granting and Revoking Privileges, Mandatory Access Control and Role-Based Access Control for Multilevel Security, SQL Injection	√		
	6		Database Security II: Introduction to Statistical Database Security, Introduction to Flow Control, Encryption and Public Key Infrastructures, Privacy Issues and Preservation, Challenges of Database Security	√		
4	7		Physical Database Design: Disk Storage, Basic File Structures, Hashing, and Modern Storage Architectures Physical Database Design Ia: Introduction, Secondary Storage Devices, Buffering Blocks, Placing File Records on Disk, Operations on Files, Files of Unordered Records (Heap Files), Files of Ordered Records (Sorted Files)	√		
	8		Data Protection RAID I	√		
5	9		Data Protection RAID II			
	10		Physical Database Design: Disk Storage, Basic File Structures, Hashing, and Modern Storage Architectures Physical Database Design Ib: Hashing Techniques, Other Primary File Organizations, Parallelizing Disk Access Using RAID Technology, Intro to New Storage systems	√		
6	11		Physical Database Design II: Indexing Structures for Files Introduction, Types of Single-Level Ordered Indexes, Multilevel Indexes, Dynamic Multilevel Indexes Using B-Trees and B+-Trees, Indexes on Multiple Keys	√		
	12		Query planning, evaluation and optimization: Introduction to Query Processing, Translating SQL Queries into Relational Algebra, Algorithms for External Sorting, Algorithms for Relational Algebra Operations, Algorithms for SELECT and JOIN Operations, Algorithms for PROJECT and SET Operations, Implementing Aggregate Operations	√		

			and Outer Joins, Combining Operations using Pipelining, Using Heuristics in Query Optimization			
7	13		Introduction to Information Storage: Information storage, Evolution of Storage Architectures, Data Center Infrastructure, Virtualization and Cloud Computing	√		
	14		Data Centre Environment: Core elements of a data center, Application, DBMS, Host, Connectivity	√		
8	15		Data Centre Environment - Storage I: Types, Disk Drive Organization, Physical Disk Structure, Disk Block Addressing	√		
	16		Data Centre Environment - Storage II: Disk Drive Performance, Host Access to Data	√		
9	17		Revision		√	
	18		CA Test		√	
10	19		Intelligent Storage Systems: Introduction, Components of an Intelligent Storage System, Storage Provisioning	√		
	20		FCSCAN: Growth of Storage Area Networks (SANs), Fibre Channel Overview, The SAN and Its Evolution, Components of FC SAN, FC Connectivity, Switched Fabric Ports, Fibre Channel Architecture, Fabric Services, Switched Fabric Login Types, Zoning, FC SAN Topologies, Virtualization in SAN	√		
11	21		IPSAN and FCoE: Introduction, iSCSI, FCIP, FCoE	√		
	22		Revision	√		
12	23		Network Attached Storage: Introduction, General-Purpose Servers versus NAS Devices, Benefits of NAS, File Systems and Network File Sharing, Components of NAS, NAS I/O Operation, NAS Implementations, NAS File-Sharing Protocols, Factors Affecting NAS Performance, File-Level Virtualization	√		
	24		Object-Based and Unified Storage: Introduction, Object-Based Storage Devices, Content-Addressed Storage, Unified Storage, Introduction to Business Continuity	√		
13	25		Revision		√	
	26		Revision		√	

Course Master: Denis L. Nkweteyim

Head of Department

NOTES: *L= Lecture, T = Tutorial, P= Practicals*

RECOMMENDED TEXTS

1. Elmasri, R. (2008). Fundamentals of database systems. Pearson Education India.
2. EMC Services (2012). Information storage and management: Storing, managing, and protecting digital information in classic, virtualized, and cloud environments. Wiley Publishing